

Notes on Chapter 20 - Spreadsheets

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Prerequisites

```
library(readxl)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(writexl)
```

Reading Excel spreadsheets using the **readxl** package:

- `read_xls()` reads Excel files with xls format.
- `read_xlsx()` reads Excel files with xlsx format.
- `read_excel()` reads both xls and xlsx formats. It infers the type based on the input.

Examples:

```
students <- read_excel("data/students.xlsx")
students
```

```
## # A tibble: 6 x 5
##   `Student ID` `Full Name`   favourite.food    mealPlan          AGE
##   <dbl> <chr>          <chr>          <chr>          <chr>
## 1         1 Sunil Huffmann Strawberry yoghurt Lunch only        4.0
## 2         2 Barclay Lynn    French fries     Lunch only        5.0
## 3         3 Jayendra Lyne   N/A             Breakfast and lunch 7.0
## 4         4 Leon Rossini      Anchovies        Lunch only        <NA>
## 5         5 Chidiegwu Dunkel    Pizza           Breakfast and lunch five
## 6         6 Güvenç Attila      Ice cream        Lunch only        6.0
```

Reading an Excel file while supplying column names at the same time:

```
read_excel(
  "data/students.xlsx",
  col_names = c("student_id", "full_name", "favorite_food", "meal_plan", "age")
)
```

```
## # A tibble: 7 x 5
##   student_id full_name      favorite_food    meal_plan          age
```

```
##   <chr>      <chr>          <chr>          <chr>          <chr>
## 1 Student ID Full Name      favourite.food    mealPlan      AGE
## 2 1.0        Sunil Huffmann Strawberry yoghurt Lunch only     4.0
## 3 2.0        Barclay Lynn   French fries   Lunch only     5.0
## 4 3.0        Jayendra Lyne  N/A           Breakfast and lunch 7.0
## 5 4.0        Leon Rossini   Anchovies     Lunch only     <NA>
## 6 5.0        Chidiegwu Dunkel Pizza          Breakfast and lunch five
## 7 6.0        Güvenç Attila  Ice cream     Lunch only     6.0
```

Reading an Excel file while supplying column names at the same time, and skipping the original header:

```
read_excel(
  "data/students.xlsx",
  col_names = c("student_id", "full_name", "favorite_food", "meal_plan", "age"),
  skip = 1
)
```

```
## # A tibble: 6 x 5
##   student_id full_name      favorite_food    meal_plan      age
##   <dbl> <chr>          <chr>          <chr>          <chr>
## 1         1 Sunil Huffmann Strawberry yoghurt Lunch only     4.0
## 2         2 Barclay Lynn   French fries   Lunch only     5.0
## 3         3 Jayendra Lyne  N/A           Breakfast and lunch 7.0
## 4         4 Leon Rossini   Anchovies     Lunch only     <NA>
## 5         5 Chidiegwu Dunkel Pizza          Breakfast and lunch five
## 6         6 Güvenç Attila  Ice cream     Lunch only     6.0
```

Reading an Excel file while supplying column names at the same time, skipping the original header, and specifying the values for NA:

```
read_excel(
  "data/students.xlsx",
  col_names = c("student_id", "full_name", "favorite_food", "meal_plan", "age"),
  skip = 1,
  na = c("", "N/A")
)
```

```
## # A tibble: 6 x 5
##   student_id full_name      favorite_food    meal_plan      age
##   <dbl> <chr>          <chr>          <chr>          <chr>
## 1         1 Sunil Huffmann Strawberry yoghurt Lunch only     4.0
## 2         2 Barclay Lynn   French fries   Lunch only     5.0
## 3         3 Jayendra Lyne  <NA>          Breakfast and lunch 7.0
## 4         4 Leon Rossini   Anchovies     Lunch only     <NA>
## 5         5 Chidiegwu Dunkel Pizza          Breakfast and lunch five
## 6         6 Güvenç Attila  Ice cream     Lunch only     6.0
```

Reading an Excel file while supplying column names at the same time, skipping the original header, and specifying the values for NA, and supplying the correct column types:

```
students <- read_excel(
  "data/students.xlsx",
  col_names = c("student_id", "full_name", "favorite_food", "meal_plan", "age"),
  skip = 1,
  na = c("", "N/A"),
  col_types = c("numeric", "text", "text", "text", "text")
)
```

```
students <- students |>
  mutate(
    age = if_else(age == "five", "5", age),
    age = parse_number(age)
  )
```

students

```
## # A tibble: 6 x 5
##   student_id full_name      favorite_food meal_plan      age
##   <dbl> <chr>          <chr>      <chr>      <dbl>
## 1         1 Sunil Huffmann Strawberry yoghurt Lunch only      4
## 2         2 Barclay Lynn   French fries   Lunch only      5
## 3         3 Jayendra Lyne   <NA>          Breakfast and lunch 7
## 4         4 Leon Rossini    Anchovies     Lunch only     NA
## 5         5 Chidiegwu Dunkel Pizza         Breakfast and lunch 5
## 6         6 Güvenç Attila   Ice cream     Lunch only      6
```

Note: we first read the `age` column as text, then corrected the issue, and then parsed it as numeric.

Reading worksheets

Specifying which “tab” to open using the `sheet` argument:

```
read_excel("data/penguins.xlsx", sheet = "Torgersen Island")
```

```
## # A tibble: 52 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <chr>   <chr>      <chr>          <chr>      <chr>          <chr>
## 1 Adelie Torgersen 39.1            18.7            181.0          3750.0
## 2 Adelie Torgersen 39.5            17.4            186.0          3800.0
## 3 Adelie Torgersen 40.3            18.0            195.0          3250.0
## 4 Adelie Torgersen NA              NA              NA              NA
## 5 Adelie Torgersen 36.7            19.3            193.0          3450.0
## 6 Adelie Torgersen 39.3            20.6            190.0          3650.0
## 7 Adelie Torgersen 38.9            17.8            181.0          3625.0
## 8 Adelie Torgersen 39.2            19.6            195.0          4675.0
## 9 Adelie Torgersen 34.1            18.1            193.0          3475.0
## 10 Adelie Torgersen 42.0            20.2            190.0          4250.0
## # i 42 more rows
## # i 2 more variables: sex <chr>, year <dbl>
```

Repeating the operation above, with the NAs specified:

```
penguins_torgersen <- read_excel(
  "data/penguins.xlsx",
  sheet = "Torgersen Island",
  na = "NA"
)
```

penguins_torgersen

```
## # A tibble: 52 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <chr>   <chr>      <dbl>          <dbl>      <dbl>          <dbl>
## 1 Adelie Torgersen 39.1            18.7            181            3750
```

```
## 2 Adelie Torgersen      39.5      17.4      186      3800
## 3 Adelie Torgersen      40.3       18      195      3250
## 4 Adelie Torgersen      NA       NA       NA       NA
## 5 Adelie Torgersen      36.7      19.3      193      3450
## 6 Adelie Torgersen      39.3      20.6      190      3650
## 7 Adelie Torgersen      38.9      17.8      181      3625
## 8 Adelie Torgersen      39.2      19.6      195      4675
## 9 Adelie Torgersen      34.1      18.1      193      3475
## 10 Adelie Torgersen     42       20.2      190      4250
## # i 42 more rows
## # i 2 more variables: sex <chr>, year <dbl>
```

What if you don't know the worksheet names (aka the tab name)? Use `excel_sheets()`.

```
excel_sheets("data/penguins.xlsx")
```

```
## [1] "Torgersen Island" "Biscoe Island"    "Dream Island"
```

Reading the remaining worksheets:

```
penguins_biscoe <- read_excel(
  "data/penguins.xlsx",
  sheet = "Biscoe Island",
  na = "NA"
)
penguins_dream <- read_excel(
  "data/penguins.xlsx",
  sheet = "Dream Island",
  na = "NA"
)
```

Checking the dimensions of each dataset extracted from the Excel worksheet:

```
dim(penguins_torgersen)
```

```
## [1] 52  8
```

```
dim(penguins_biscoe)
```

```
## [1] 168  8
```

```
dim(penguins_dream)
```

```
## [1] 124  8
```

Combining the files using `bind_rows()`:

```
penguins <- bind_rows(
  penguins_torgersen,
  penguins_biscoe,
  penguins_dream
)
```

```
penguins
```

```
## # A tibble: 344 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <chr>    <chr>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 Adelie Torgersen      39.1           18.7           181           3750
## 2 Adelie Torgersen      39.5           17.4           186           3800
```

```
## 3 Adelie Torgersen      40.3      18      195      3250
## 4 Adelie Torgersen      NA       NA       NA       NA
## 5 Adelie Torgersen      36.7     19.3     193     3450
## 6 Adelie Torgersen      39.3     20.6     190     3650
## 7 Adelie Torgersen      38.9     17.8     181     3625
## 8 Adelie Torgersen      39.2     19.6     195     4675
## 9 Adelie Torgersen      34.1     18.1     193     3475
## 10 Adelie Torgersen     42       20.2     190     4250
## # i 334 more rows
## # i 2 more variables: sex <chr>, year <dbl>
```

Reading part of a sheet

Using `readxl_example()` to locate the spreadsheet in the directory: `deaths_path <- readxl_example("death.xlsx")`

Reading excel while specifying the data range: `read_excel(deaths_path, range = "A5:F15")`

Writing to excel

Using `write_xlsx()`:

```
bake_sale <- tibble(
  item = factor(c("brownie", "cupcake", "cookie")),
  quantity = c(10, 5, 8)
)
```

```
bake_sale
```

```
## # A tibble: 3 x 2
##   item      quantity
##   <fct>      <dbl>
## 1 brownie      10
## 2 cupcake       5
## 3 cookie       8
```

```
write_xlsx(bake_sale, path = "data/bake-sale.xlsx")
```

To create an Excel spreadsheet with more sophisticated formatting, we can use the **openxlsx** package. This package, however, is not part of tidyverse.

Reading Google Sheets

Prerequisites

```
library(google sheets4)
library(tidyverse)
```

We can use either the URL or the google sheet ID: `gs4_deauth()`

```
gs4_deauth()
students_sheet_id <- "1V1nPp1tz0uutXFLb3G9Eyx13qxeEhn0XUzL5_BcCQ0w"

students <- read_sheet(students_sheet_id)
```

```
## v Reading from "students".
```

```
## v Range 'Sheet1'.
```

```
students
```

```
## # A tibble: 6 x 5
##   `Student ID` `Full Name`      favourite.food      mealPlan      AGE
##   <dbl> <chr>          <chr>          <chr>          <list>
## 1         1 Sunil Huffmann Strawberry yoghurt Lunch only      <dbl [1]>
## 2         2 Barclay Lynn   French fries    Lunch only      <dbl [1]>
## 3         3 Jayendra Lyne  N/A            Breakfast and lunch <dbl [1]>
## 4         4 Leon Rossini    Anchovies      Lunch only      <NULL>
## 5         5 Chidiegwu Dunkel Pizza          Breakfast and lunch <chr [1]>
## 6         6 Güvenç Attila    Ice cream      Lunch only      <dbl [1]>
```

Specifying the column names, NA stings, and column types while reading the spreadsheet:

```
students <- read_sheet(
  students_sheet_id,
  col_names = c("student_id", "full_name", "favorite_food", "meal_plan", "age"),
  skip = 1,
  na = c("", "N/A"),
  col_types = "dcccc"
)
```

```
## v Reading from "students".
```

```
## v Range '2:10000000'.
```

```
students
```

```
## # A tibble: 6 x 5
##   student_id full_name      favorite_food      meal_plan      age
##   <dbl> <chr>          <chr>          <chr>          <chr>
## 1         1 Sunil Huffmann Strawberry yoghurt Lunch only      4
## 2         2 Barclay Lynn   French fries    Lunch only      5
## 3         3 Jayendra Lyne  <NA>           Breakfast and lunch 7
## 4         4 Leon Rossini    Anchovies      Lunch only      <NA>
## 5         5 Chidiegwu Dunkel Pizza          Breakfast and lunch five
## 6         6 Güvenç Attila    Ice cream      Lunch only      6
```

Example using the penguin dataset:

```
penguins_sheet_id <- "1aFu8lnD_g0yjfF50-K6SFgSEWiHPpgvFCFONY9D6LXnY"
read_sheet(penguins_sheet_id, sheet = "Torgersen Island")
```

```
## v Reading from "penguins".
```

```
## v Range "'Torgersen Island'".
```

```
## # A tibble: 52 x 8
##   species island      bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <chr>   <chr>      <list>          <list>          <list>          <list>
## 1 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 2 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 3 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 4 Adelie Torgersen <chr [1]>       <chr [1]>       <chr [1]>       <chr [1]>
## 5 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 6 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 7 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 8 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## 9 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
```

```
## 10 Adelie Torgersen <dbl [1]>      <dbl [1]>      <dbl [1]>      <dbl [1]>
## # i 42 more rows
## # i 2 more variables: sex <chr>, year <dbl>
```

Worksheet names can be listed using `sheet_names()`:

```
sheet_names(penguins_sheet_id)
```

```
## [1] "Torgersen Island" "Biscoe Island"   "Dream Island"
```

Writing to Google Sheets

Basic command:

```
write_sheet(bake_sale, ss = "bake-sale")
```

Writing data to a specific worksheet inside a Google sheet:

```
write_sheet(bake_sale, ss = "bake-sale", sheet = "Sales")
```

Reading from a public Google sheet without authentication with your Google account: `gs4_deauth()`

When you attempt to read in a sheet that requires authentication, googlesheets4 will direct you to a web browser with a prompt to sign in.

To use an authentication token instead, the command is: `gs4_auth(email = "xyz@google.com")`